

- This world has shifted now.
- Web and mobile applications require instant feedback.
 - The Internet of Things (IoT) is sending terabytes of data every second, 24/7.
 - Stock trading systems need decision making in sub-second timeframes.

Because of this, data integration has turned from daily jobs to always-on streaming.

Batch vs stream processing

Information not only supports decisions, it supports timing. The decision to process in streams or to use batch

processing can alter the speed at which your business reacts and changes. Both options are compared in Figure 2 and Table 1.

In Figure 3 we can see the business scenario and the recommended approach. It also depicts the reason for selecting this approach.

Instead of thinking of it as an ‘either-or’ decision, think of streaming and batch as additional tools in your toolbox or instrument panel. In fact, many modern systems use a hybrid of the two:

- Batch to drive analytics, machine learning and data lakes
- Stream to drive responsiveness, customer interaction and system intelligence

The most agile businesses today leverage the depth of batch against the fast-paced speed of streaming, and that is where the real win is.

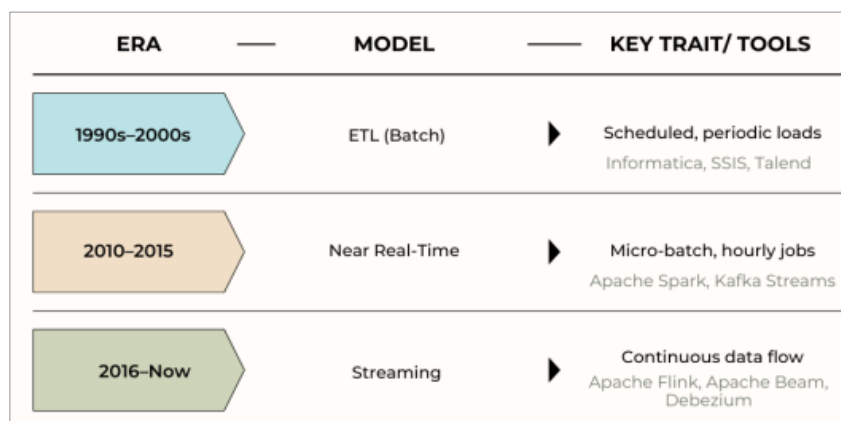


Figure 1: Timeline of evolution

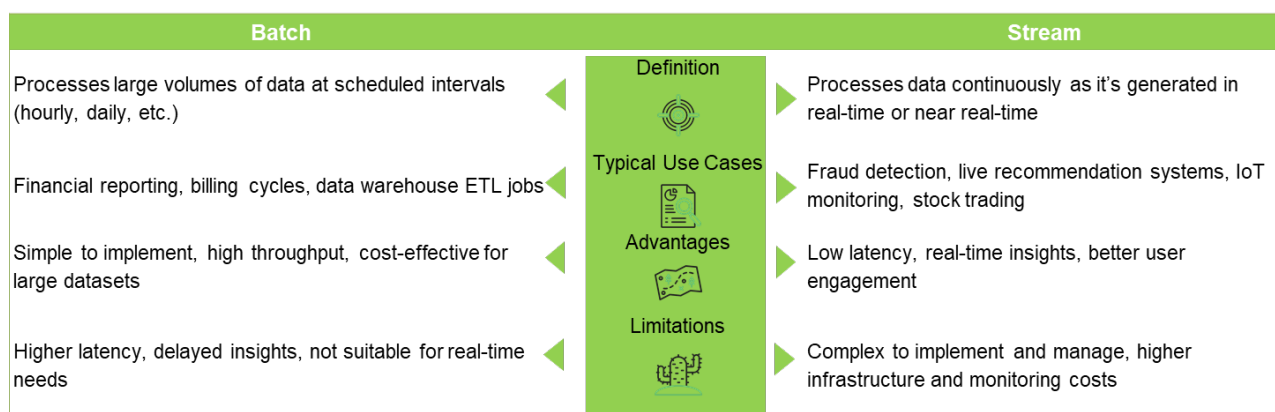


Figure 2: Stream vs batch processing: Feature comparison

Table 1: Stream and batch processing: Feature comparison

Feature	Batch processing	Stream processing
Latency	Minutes to hours	Milliseconds to seconds
Data input style	Periodic (batches)	Continuous (events or messages)
Use case focus	Historical/trend analysis	Instantaneous insights
Infrastructure need	Moderate	High-performance, always-on systems
Complexity	Simpler logic and orchestration	Complex with state management, windowing, etc
Operational cost	Lower for offline workloads	Higher for 24/7 responsiveness
Best for	Data warehousing, reports, analytics	Monitoring, alerts, live dashboards